Personal Expense Tracker

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Overview

The Personal Expense Tracker is a desktop application for recording, organising and reviewing personal income and expenses. It brings transaction entry, summaries, charts and reports into one program, replacing scattered notes and manual arithmetic with structured, searchable records.

Problem and Aim

People who track money by hand often end up with incomplete records, inconsistent categories and calculation errors, and find it hard to see where their money goes. The aim of this project is to provide a simple local application that centralises income and expense records and makes monthly review easy through totals, charts and filtered reports.

Key Features

- **Transaction management:** add, view, edit and delete income and expense records, each with date, category, amount, payment method and description.
- **Dashboard:** total income, total expenses, balance, current-month spending, savings rate, a monthly summary and recent transactions.
- **Charts:** expense distribution by category, monthly income versus expense, and monthly expense trend.
- **Reports:** filter by date range, transaction type, category and search text, with CSV and Excel export.

Technology Stack

- **Python** for application logic
- **Tkinter** for the graphical interface
- **SQLite** for local, persistent storage (a single `transactions` table)

- **Matplotlib** for charts
- **OpenPyXL** for Excel workbook support

Design

The application is modular, with separate files for the dashboard, expenses, income, charts, reports and database operations. Interface screens call shared database functions that use parameterised SQL, which keeps data handling consistent and reduces SQL-injection risk.

Current Status and Limitations

The application is a working academic prototype with all main screens implemented. Testing is partially complete, and a full test plan of 14 cases is ready to be run. It is a local, single-user tool: there is no login, cloud sync or bank integration.

Future Work

Unusual-spending detection, expense forecasting and budget recommendations; backup and restore; stronger input validation and automated tests; bank-statement import; and packaging as an installer.

Learning Outcomes

The project combines GUI programming, relational databases and SQL, data aggregation, visualisation and software testing, and shows the value of modular design and careful handling of financial data.